BKT Transition: 0/0 Without Symmetry Breaking

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The Removable Singularity Framework

Abstract

We show that the Berezinskii-Kosterlitz-Thouless (BKT) transition is a 0/0 removable singularity that occurs WITHOUT any symmetry breaking -- fundamentally different from Ising. The 2D XY model exhibits a topological phase transition where vortex-antivortex pairs unbind at T_BKT. The universal jump eta(T_BKT) = 1/4 EXACTLY and K_c = 2/pi provide exact 0/0 signatures. The order parameter is ALWAYS zero -- there is no symmetry breaking. This introduces a FOURTH universality class: topological (eta=1/4), distinct from Ising (beta=1/8), Kolmogorov (-5/3), and quantum (z=1). Nobel Prize 2016 (Thouless, Haldane, Kosterlitz).

1. The 2D XY Model

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H = -J * sum(cos(theta_i - theta_j))
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The 2D XY model has U(1) symmetry (continuous rotation). Spins are 2D angles theta_i. Unlike Ising (Z2 symmetry), the continuous symmetry forbids long-range order in 2D (Mermin-Wagner theorem). But the BKT transition creates quasi-long-range order with power-law correlations.

2. Correlation Function

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G(r) = <cos(theta_i - theta_j)> ~ r^{-eta(T)}
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Below T_BKT: G(r) decays as a power law r^{-eta} with eta < 1/4. This is quasi-long-range order -- correlations decay algebraically but never fully vanish. Above T_BKT: G(r) decays exponentially with correlation length xi. At T_BKT: eta = 1/4 EXACTLY -- the universal jump.

3. Universal Jump: eta = 1/4

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eta(T_BKT) = 1/4  EXACTLY  (Berezinskii 1971)
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The most remarkable result: at T_BKT, the exponent eta jumps DISCONTINUOUSLY from 1/4 to larger values. This is a removable singularity in the exponent. The stiffness jumps: K_c = 2/pi. But the energy and specific heat are CONTINUOUS -- the transition is invisible in thermodynamics!

4. Vortex-Antivortex Unbinding

The mechanism is topological: vortices are point defects with winding number +/-1. Below T_BKT: vortices are bound in vortex-antivortex pairs (free energy F_v > 0). Above T_BKT: vortices unbind and proliferate (F_v < 0). At T_BKT: 0/0 -- the binding/unbinding transition.

- Vortex energy: E_v = pi*J (cost to create a vortex)
- Vortex entropy: S_v = 2*pi*ln(L) (logarithmic)
- Free energy: F_v = E_v - T*S_v = pi*J - 2*pi*T*ln(L)
- At T_BKT: F_v = 0 (0/0 removable singularity)

5. Kosterlitz RG Flow

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dy/dl = (2 - pi*K)*y,  dK/dl = -pi^2 * y^2 * K^2
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The renormalization group flow has two regimes. Below T_BKT: y flows to 0 (vortex fugacity vanishes, bound pairs). Above T_BKT: y flows to infinity (vortices proliferate). At T_BKT: the flow changes direction (0/0 in the RG equations). The fixed point (K_c, 0) is the BKT critical point.

6. Four Universality Classes

The BKT transition introduces a FOURTH distinct universality class in our framework:

- Ising (Ch.36): symmetry breaking, beta=1/8, Z2
- Kolmogorov (Ch.37): cascade, -5/3, translation
- Quantum (Ch.39): quantum fluctuations, beta=1/8, z=1
- BKT (Ch.40): topological, eta=1/4, U(1), NO symmetry breaking

BKT is fundamentally different: the order parameter is ALWAYS zero. There is no symmetry breaking. The transition is purely topological.

7. Nobel Prize 2016

The Nobel Prize in Physics 2016 was awarded to David J. Thouless, F. Duncan M. Haldane, and J. Michael Kosterlitz "for theoretical discoveries of topological phase transitions and topological phases of matter." The BKT transition was the first example of a topological phase transition.

8. Connections to Prior 0/0 Singularities

BKT	Prior Chapter	Connection
eta=1/4	Ch.36 Ising	Different universality
No symmetry breaking	Ch.39 Quantum	No 1D quantum maps
Vortex unbinding	Ch.37 Turbulence	Topological defects
Topological order	Ch.24 E8	Lie algebras classify
Power-law correlations	Ch.34 Consciousness	Critical brain

9. Conclusion

The BKT transition is the most exotic 0/0 removable singularity. It occurs WITHOUT any symmetry breaking -- the order parameter is always zero. The universal jump eta = 1/4 and K_c = 2/pi are exact signatures. The vortex-antivortex unbinding mechanism is purely topological. This introduces a fourth universality class (topological, eta=1/4) distinct from Ising, Kolmogorov, and quantum. The BKT 0/0 reveals that phase transitions can occur through purely topological mechanisms, without any change in symmetry.

References

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